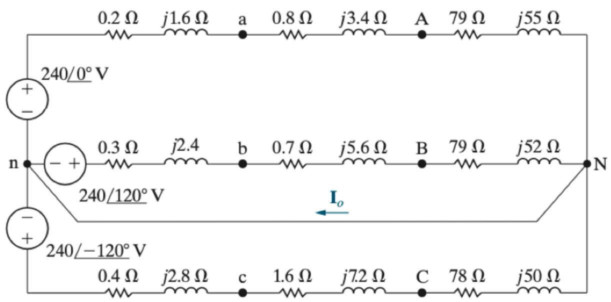
11.6 a) Find $\mathbf{I}_o$ in the circuit in Fig. P11.6.

PSPICE MULTISIM b) Find $\mathbf{V}_{AN}$.

c) Find $\mathbf{V}_{AB}$.

d) Is the circuit a balanced or unbalanced three-phase system?

Figure P11.6



11.10 The phase voltage at the terminals of a balanced three-phase Y-connected load is 2400 V. The load has an impedance of $16 + j12\ \Omega/\phi$ and is fed from a line having an impedance of $0.10 + j0.80\ \Omega/\phi$. The Y-connected source at the sending end of the line has a phase sequence of acb and an internal impedance of $0.02 + j0.16\ \Omega/\phi$. Use the a-phase voltage at the load as the reference and calculate (a) the line currents $\mathbf{I}_{aA}$, $\mathbf{I}_{bB}$, and $\mathbf{I}_{cC}$; (b) the line voltages at the source, $\mathbf{V}_{ab}$, $\mathbf{V}_{bc}$ and $\mathbf{V}_{ca}$; and (c) the internal phase-to-neutral voltages at the source, $\mathbf{V}_{a'n}$, $\mathbf{V}_{b'n}$, and $\mathbf{V}_{c'n}$.

11.15 A balanced Y-connected load having an impedance of $96 - j28\ \Omega/\phi$ is connected in parallel with a balanced Δ-connected load having an impedance of $144 + j42\ \Omega/\phi$. The parallel loads are fed from a line having an impedance of $j1.5\ \Omega/\phi$. The magnitude of the phase voltage of the Y-load is 7500 V.

a) Calculate the magnitude of the phase current in the Y-connected load.

b) Calculate the magnitude of the phase current in the Δ-connected load.

c) Calculate the magnitude of the current in the line feeding the loads.

d) Calculate the magnitude of the line voltage at the sending end of the line.

11.19 A three-phase Δ-connected generator has an internal impedance of $0.6 + j4.8\ \Omega/\phi$. When the load is removed from the generator, the magnitude of the terminal voltage is 34,500 V. The generator feeds a Δ-connected load through a transmission line with an impedance of $0.8 + j6.4\ \Omega/\phi$. The per-phase impedance of the load is $2877 - j864\ \Omega$.

a) Construct a single-phase equivalent circuit.

b) Calculate the magnitude of the line current.

c) Calculate the magnitude of the line voltage at the terminals of the load.

d) Calculate the magnitude of the line voltage at the terminals of the source.

e) Calculate the magnitude of the phase current in the load.

f) Calculate the magnitude of the phase current in the source.

11.24 In a balanced three-phase system, the source has an abc sequence, is Y-connected, and $\mathbf{V}_{an} = 250\angle{-60^\circ}$ V. The source feeds two loads, both of which are Y-connected. The impedance of load 1 is $15 + j20\ \Omega/\phi$. The complex power for the a-phase of load 2 is $500\angle{45^\circ}$ VA. Find the total complex power supplied by the source.

11.26 A three-phase positive sequence Y-connected source supplies 24 kVA with a power factor of 0.6 lagging to a parallel combination of a Y-connected load and a Δ-connected load. The Y-connected load uses 12 kVA at a power factor of 0.8 lagging and has an a-phase current of $40\angle 60^\circ$ A.

a) Find the complex power per phase of the Δ-connected load.

b) Find the magnitude of the line voltage.